iwinfo

Release 5.0.0

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CONTENTS

CHAPTER

ONE

IWINFO

1.1 Overview

iwinfo : Provide useful information about wireless network(s)

This is a command line program which is run in any terminal. It shows some information about existing wireless connections along with the result of an optional scan for wireless devices. Scan is turned on using *-s* option.

Scanning wireless networks requires elevated privileges, which means either by running as root or by being provided with the required cap_net_xxx capablilities. Scanning is only permitted to users who are members of *wheel* group.

This package provides the application, written in python, along with a small C-program which is installed with cap_net_raw and cap_net_admin[2] and it provides the capabilities for the program to be run non-root.

Since this does add some level of risk, scanning is limited to root and members of the *wheel* group only.

Some addititional information is gathered using *iwctl* when *iwd* is running. These are the *ip addres*, *security mode* (e.g. WPA2), and *transmission type* (e.g. 802.11ax). These require either root or membership in *network* or *wheel* groups.

Others will still be able to get local wireless device and connection info, but will not be able to scan the network(s).

Summary:

Info	User	Group *network*	Group *wheel* or root
Basic	✓	✓	✓
Extra	×	✓	✓
Scan	×	×	✓

- All git tags are signed with arch@sapience.com key which is available via WKD or download from https://www.sapience.com/tech. Add the key to your package builder gpg keyring. The key is included in the Arch package and the source= line with *?signed* at the end can be used to verify the git tag. You can also manually verify the signature

1.2 Key features

- Shows local wireless device(s) and connection info.
- Show summary of wireless hardware capabilities
- Scans wireless network(s) and provides compact report

[2] See man capabilities.

1.3 New / Interesting

5.0.0

- Reorg source a little
- Switch python packaging from hatch to uv
- License GPL-2.0-or-later

4.8.0

- New verbose option *-v* reports all frequencies supported by each phy

  This will also mark any disabled channels (restricted by reg domain). It will show if a channel is connect only using *no initiate radio*). i.e. any no initiate phy cannot be run as an Access Point. DFS channels may display *radar detected*

4.7.0

- Bug fix: Scan occasionally gets device busy which is handled by a couple of retries. However, when device is not responding at all this kept trying repeatedly. Fixed
- Bug fix: Misidentification of some phy(sical) (hardware) capabilities. Notably 802.11be (wifi-7)

Older

- Code now complies with: PEP-8, PEP-257, PEP-484 and PEP-561
- Code Refactor & clean ups.
- Wireless "host" database file name.

  Preferred name is now known-hosts.toml, which aligns better with it's purpose and format. The previous names will continue to work just fine as well.

  The known host file will first be looked for in the directory *./etc/iwinfo/* and then */etc/iwinfo/*.

CHAPTER

TWO

GETTING STARTED

2.1 Usage

Run in a terminal :

```
iwinfo --help
iwinfo
iwinfo --scan
```

2.2 Configuration

An optional configuration file for iwinfo goes in:

```
/etc/iwinfo/wifi.db
```

wifi.db allows you to provide additional information about known wireless devices on the netwwork. File is in *toml* format and a sample is installed in */etc/iwinfo/wifi.db.sample*. If available, then this information is used in generating the reports.

Each device listed in the file should have an entry of the form:

```
[ap0]
    ip = '10.0.0.10'
    mac_map = [['5GHz', 'x:x:x:x:x:x'],
           ['24Ghz', 'x:x:x:x:x:x'],
            ['lan', 'x:x:x:x:x:x'],
            ]
    model = 'Netgear R9000'
    info = 'Location Office 1'
```

The mac_map is a list of pairs of [key, mac-address]. The key can be any convenient string you choose.

2.2.1 Options

By default no network scan is performed. To turn this on use:

- (*-s, –scan*)

2.2.2 Sample Output

Sample output:

```
Interfaces:
  wlan0:
      ap_bssid : xx:xx:xx:xx:xx:xx : Netgear xr500 Location Office 1
          ssid :  MagicalPlaces
          freq :  5745.0
      signal :  -53 dBm
  rx_bitrate :  866.7 MBit/s VHT-MCS 9 80MHz short GI VHT-NSS 2
  tx_bitrate :  866.7 MBit/s VHT-MCS 9 80MHz short GI VHT-NSS 2

  Devices:
      phy0:
              wifi-6E (802.11ax)   3-bands : 2.4-GHz 5-GHz 6-GHz
```

With –scan:

```
Scan Results:
  wlan0:
  xx:xx:xx:xx:xx:xx:  MagicalPlaces-24     2432.0   -32.00 dBm : Netgear 9000  Office␣
↪1
* xx:xx:xx:xx:xx:xx:  MagicalPlaces        5745.0   -49.00 dBm : Netgear 9000  Office␣
↪1
  yy:yy:yy:yy:yy:yy:  MyNeighbor-6G        5955.0   -55.00 dBm : Asus GT11000  Test␣
↪Lab
  ...
```

The asterisk indicates machine is currently connected to that AP

CHAPTER

THREE

APPENDIX

3.1 Note on CET Shadow Stack

The code is compiled with this turned on. If for some reason you get an error compiling then you may turn it off by changing the load flag to 'cet-report=warning' in *src/ambient/Makefile*.

This may happen if you have old glibc (pre 2.39).

3.2 Installation

Available on

- Github
- Archlinux AUR

On Arch you can build using the provided PKGBUILD in the packaging directory or from the AUR. To build manually, clone the repo and :

```
rm -f dist/*
/usr/bin/python -m build --wheel --no-isolation
root_dest="/"
./scripts/do-install $root_dest
```

When running as non-root then set root_dest a user writable directory

3.3 Dependencies

- Run Time :
  - python (>= 3.11)
- Building Package:
  - git
  - hatch
  - wheel
  - build
  - installer
  - rsync
  - gcc

  - make
  - libcap-ng
- Optional to buil docs:
  - sphinx
  - texlive-latexextra (archlinux packaguing of texlive tools)

3.4 Philosophy

We follow the *live at head commit* philosophy as recommended by Google's Abseil team[1]. This means we recommend using the latest commit on git master branch.

3.5 License

Created by Gene C. and licensed under the terms of the GPL-2.0-or-later license.

- SPDX-License-Identifier: GPL-2.0-or-later
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[1] https://abseil.io/about/philosophy#upgrade-support

CHAPTER FOUR

LICENSE

iwinfo: Provide wifi information about capabilities and network(s)